



北京大学
PEKING UNIVERSITY

《集成电路工程算法》课程大作业 ——基于图建模与MILP的数据流调度

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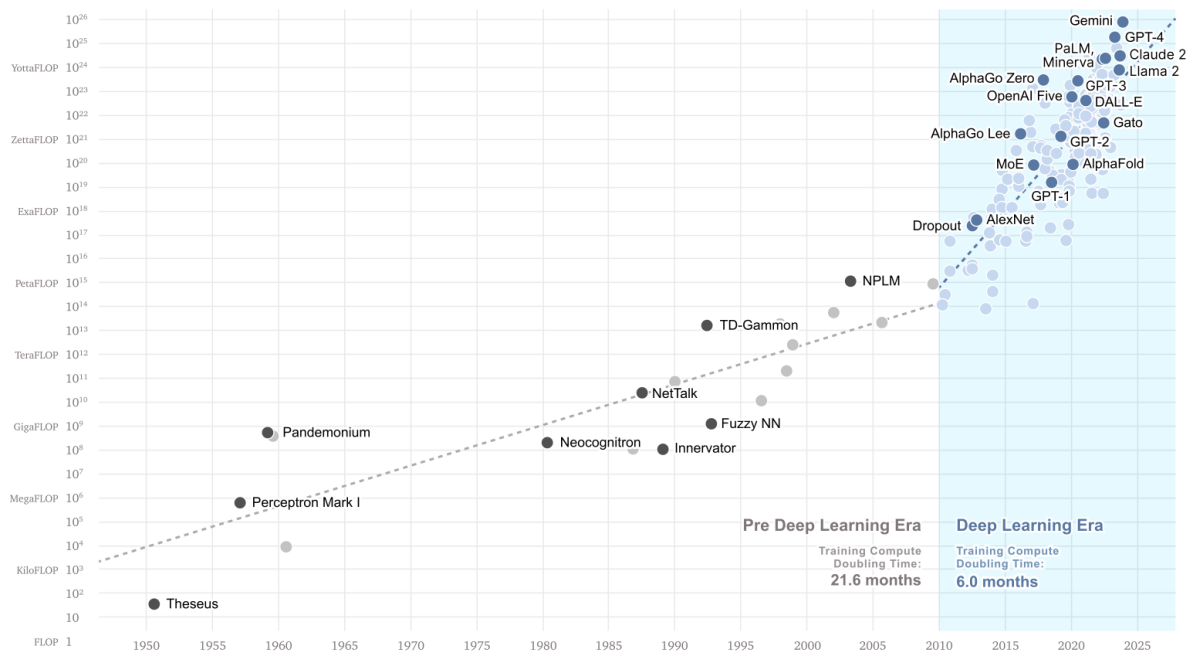
Background

❑ Fast-growing requirement

❑ Emerging hardware architecture

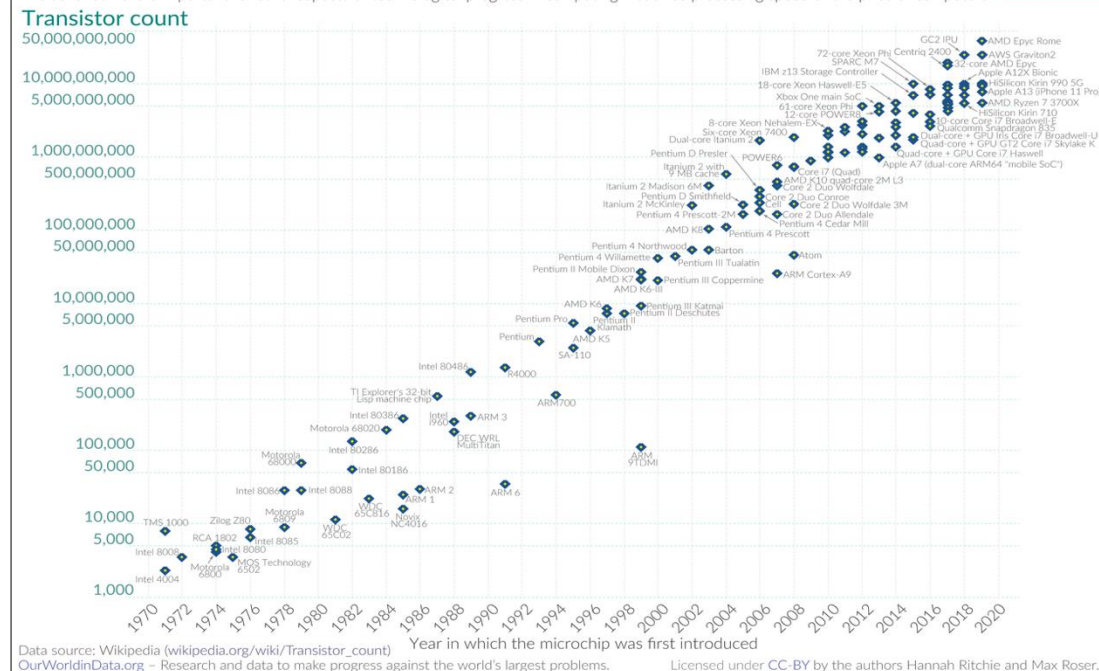
Compute Used for AI Training Runs

Total compute used to train notable AI models, measured in total FLOP (floating-point operations) | Logarithmic



Moore's Law: The number of transistors on microchips doubles approximately every two years

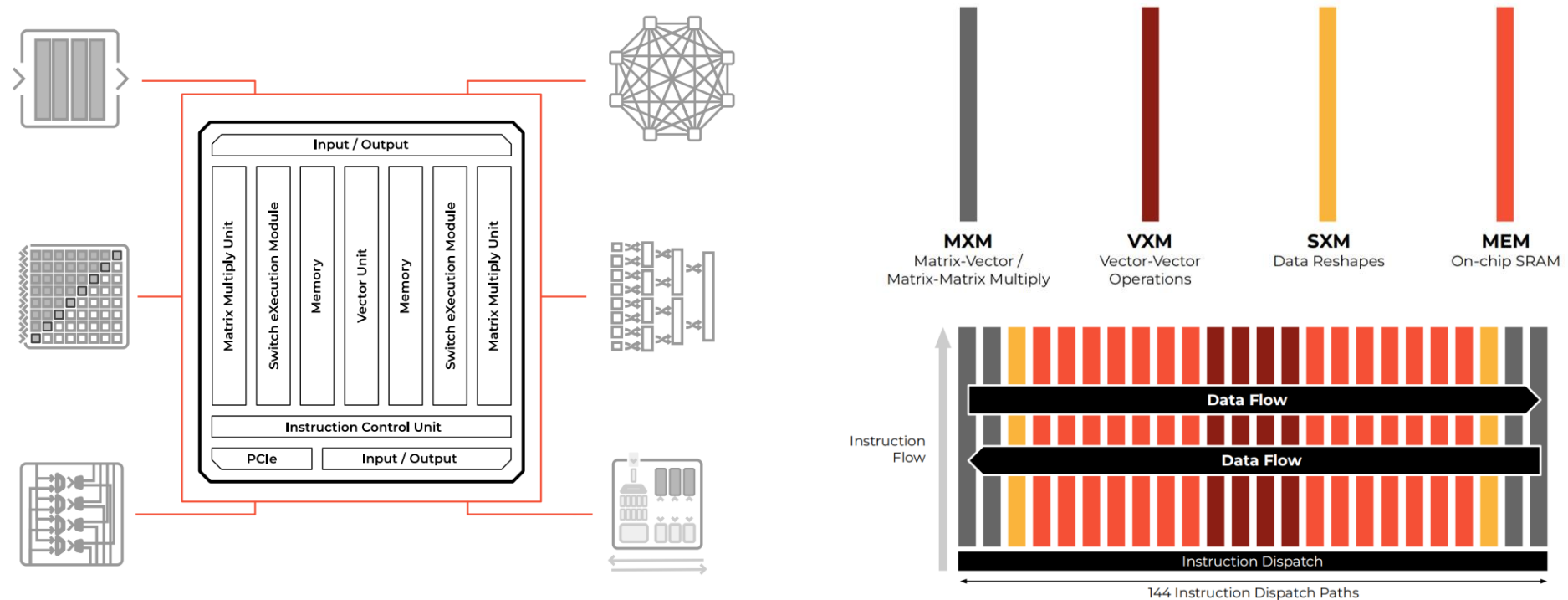
Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.



Background

□ Tensor Streaming Processor (TSP)

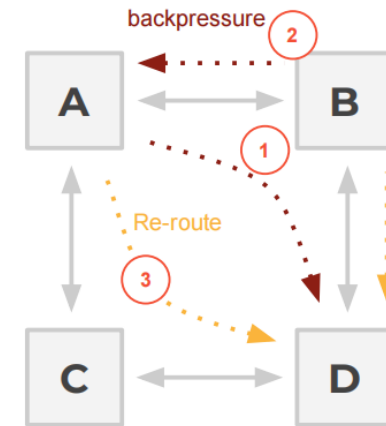
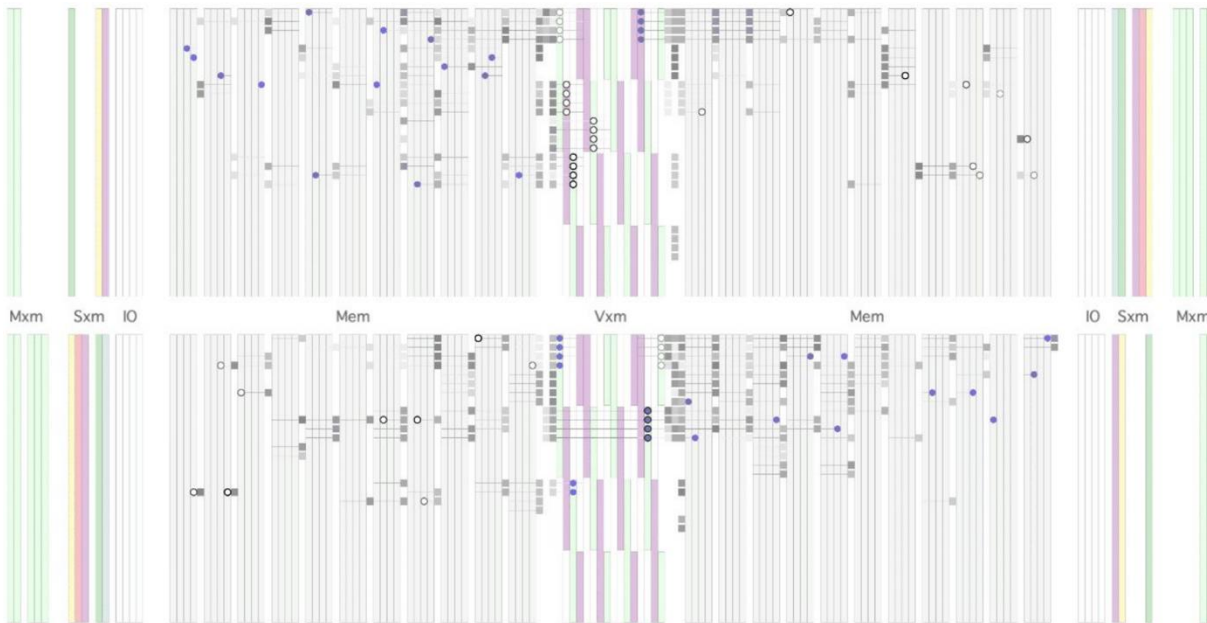
- No dynamic hardware caching: Compiler aware of all data locations at any given point in time
- Functionally sliced microarchitecture: MXM+SXM+MEM+VXM+MEM+SXM+MXM
- Simplified dataflow through streaming registers



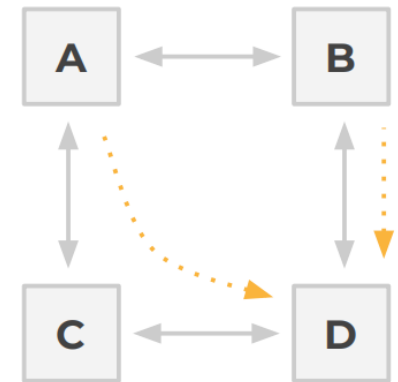
Motivation

□ Spatial-Temporal Intersection

- Conventional scheduling: unpredictable latency, network backpressure, ...
- TSP scheduling: avoid congestion, deterministic routing & execution
- Cycle-accurate instruction scheduling



**Traditional
Non-deterministic
Network**

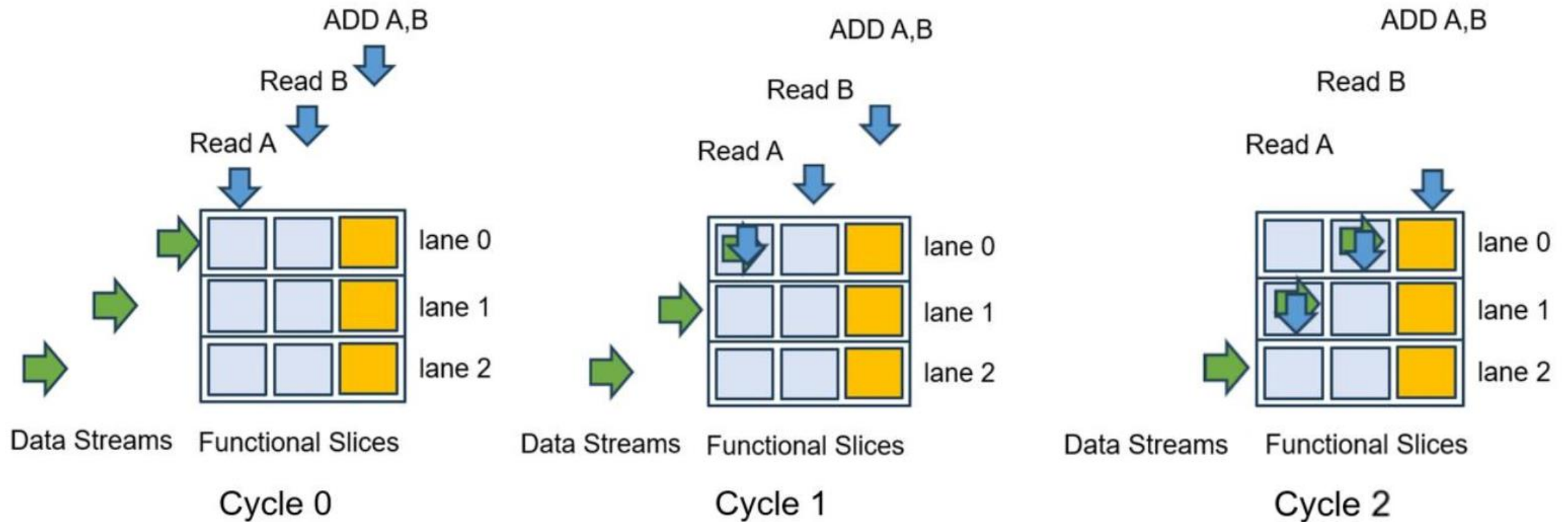


**Software-scheduled
Network**

Problem Formulation

□ Accurate data-instruction rendezvous

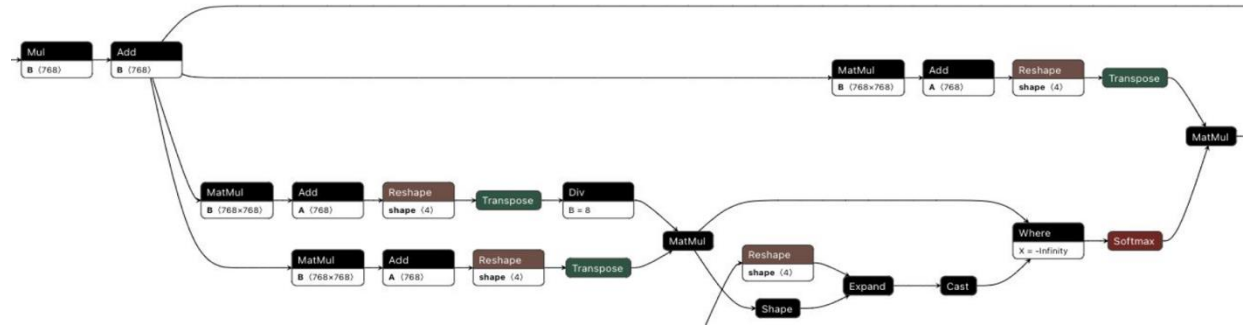
- Instruction flow: vertical flow issued by ICU
- Data flow: horizontal flow by streaming register



Problem Formulation

□ Problem Input:

- Compute graph (DAG)
- User requirement (Optional)



□ Problem Output:

- Physical mapped slice block of operator v
- Instruction's issue time of operator v



□ Optimization Object:

$$\min_{loc(v), t_{issue}(v)} \max_{v \in V} t_{issue}(v) + t_{skew}(v) + d_{func}(v)$$

Problem Formulation

□ Decision Variables:

$t_{issue}(v) \in \mathbb{N}^+$: Instruction issue time of operator v

$x_{v,s} \in \{0,1\}$: Indicate whether operator v is mapped on slice s

□ Constraints:

$$t_{issue}(v) + t_{skew}(v) = t_{issue}(u) + d_{func}(u) + Dist(loc(u), loc(v)) \quad \text{Issue time alignment}$$

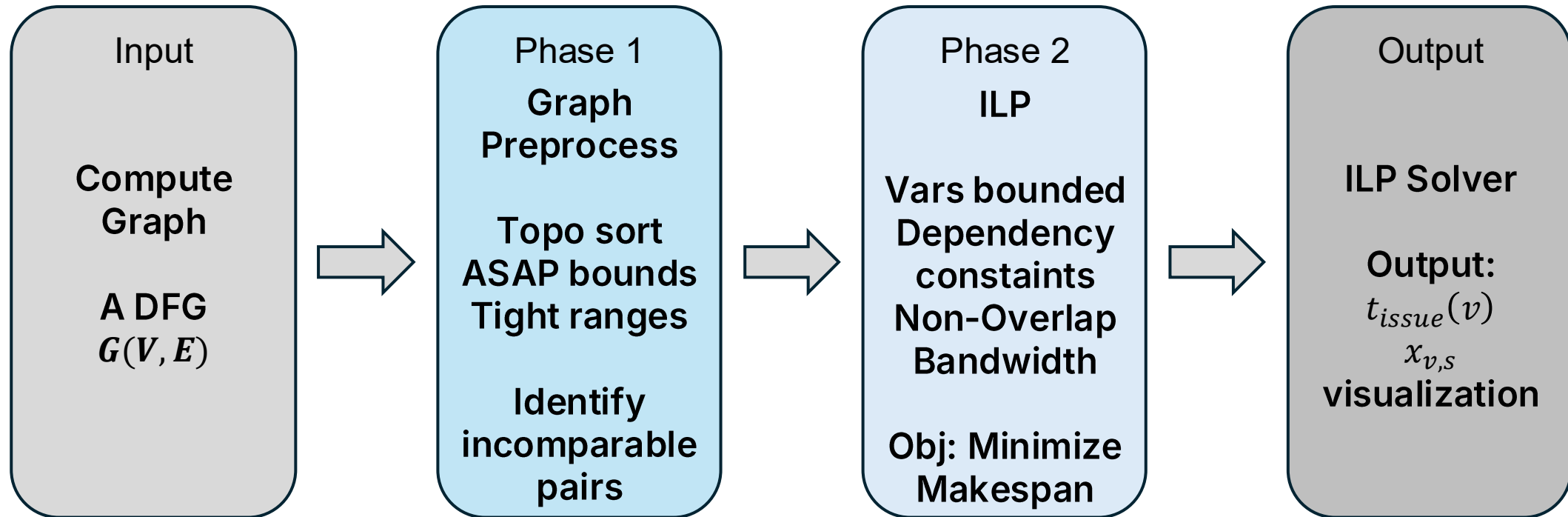
$$\sum_{s \in S} x_{v,s} = 1, \forall v \in V \quad x_{v,s} = 0, \forall v \in V, \forall s \in S \text{ if } type(v) \neq type(s) \quad \text{Spatial Assignment}$$

$$\begin{aligned} t_{issue}(u) + t_{skew}(u) + d_{func}(u) &\leq t_{issue}(v) + t_{skew}(v) + \mathcal{M} \cdot (3 - x_{u,s} - x_{v,s} - [u \rightarrow v]) \\ t_{issue}(v) + t_{skew}(v) + d_{func}(v) &\leq t_{issue}(u) + t_{skew}(u) + \mathcal{M} \cdot (2 - x_{u,s} - x_{v,s} + [u \rightarrow v]) \end{aligned} \quad \text{Non-overlapping}$$

$$\sum_{e \in E} ActiveFlow(e, s, t) \leq \mathbb{C}_{stream}, \forall s \in S, \forall t \quad \text{Stream bandwidth}$$

Solution

□ 2-Phase Approach:



Solution

□ Topological Analysis:

- Perform **topological sort** on the DFG
- Compute **ASAP** (As Soon As Possible) for each node v :
$$ASAP(v) = \max_{u: u \rightarrow v} (ASAP(u) + d_{func}(u) + Dist(loc(u), loc(v)))$$
- Compute **ALAP** (As Late As Possible) for each node v :
$$ALAP(v) = \min_{w: v \rightarrow w} (ALAP(w) - d_{func}(v) - Dist(loc(v), loc(w)))$$
- Tighten variable bounds: $ASAP(v) \leq t_{issue}(v) \leq ALAP(v)$

□ Conflict Pair Reduction:

- Compute **transitive closure** of DFG
- For two operators u, v : if $u \rightsquigarrow v$ or $v \rightsquigarrow u$ exists $\rightarrow$ ordering already determined by dependency $\rightarrow$ **skip non-overlapping constraint**
- Only generate non-overlapping constraints for **incomparable pairs**:
$$\mathcal{P} = \{(u, v) \mid u \not\rightsquigarrow v \text{ and } v \not\rightsquigarrow u\}$$
- Constraint reduction: from $O(|V|^2 \cdot |S|)$ $\rightarrow$ significantly fewer

Solution

□ Decision Variables (tightened):

$$ASAP(v) \leq t_{issue}(v) \leq ALAP(v), t_{issue}(v) \in \mathbb{N}^+ x_{v,s} \in \{0, 1\}, \forall v \in V, s \in S$$

□ Objective:

$$\min T_{makespan} = \max_{v \in V_{sink}} (t_{issue}(v) + d_{func}(v))$$

□ Constraints (inherited from problem + graph-reduced):

- **Dependency** (from DAG edges, no Big-M needed):

$$t_{issue}(v) + t_{skew}(v) = t_{issue}(u) + d_{func}(u) + Dist(loc(u), loc(v)), \forall (u, v) \in E$$

- **Spatial Assignment:**

$$\sum_{s \in S} x_{v,s} = 1, \forall v \in V x_{v,s} = 0, \text{ if } type(v) \neq type(s)$$

- **Non-overlapping** (only for $(u, v) \in \mathcal{P}$, same slice):

$$t_{issue}(u) + t_{skew}(u) + d_{func}(u) \leq t_{issue}(v) + t_{skew}(v) + \mathcal{M}(3 - x_{u,s} - x_{v,s} - [u \rightarrow v])$$

$$t_{issue}(v) + t_{skew}(v) + d_{func}(v) \leq t_{issue}(u) + t_{skew}(u) + \mathcal{M}(2 - x_{u,s} - x_{v,s} + [u \rightarrow v])$$

- **Stream Bandwidth:**

$$\sum_{e \in E} ActiveFlow(e, s, t) \leq \mathbb{C}_{stream}, \forall s \in S, \forall t$$

Evaluation Method

Metric	Definition	Goal
Makespan	$T_{makespan} = \max_{v \in V_{sink}} (t_{issue}(v) + d_{func}(v))$	Minimize
Constraint Satisfaction	All 4 constraints verified post-solve	Must = 100%
Variable Bound Tightness	$\frac{ALAP(v) - ASAP(v)}{T_{makespan}}$ averaged over v	Smaller = better preprocessing
Constraint Reduction Rate	$1 - \frac{ \mathcal{P} }{ V ^2}$	Maximize
Solve Time	Wall-clock time of ILP solver	Minimize

$$\text{Improvement} = \frac{T_{makespan}^{pureILP} - T_{makespan}^{graph+ILP}}{T_{makespan}^{pureILP}} \times 100\%$$

Method	Makespan	Solve Time	Constraints
Pure ILP (baseline)	T_1	t_1	$O(\ V\ ^2 \cdot \ S\)$
Graph + ILP (ours)	T_2	t_2	$O(\ \mathcal{P}\ \cdot \ S\)$

Expected: $T_1 = T_2$ and $t_2 \ll t_1$

Expected Output & Timeline

Aspect	Expectation
Solution Quality	Optimal or near-optimal makespan
Constraint Reduction	$> 30\%$ fewer non-overlapping constraints via graph pruning
Speedup	$> 2 \times$ faster solve time vs pure ILP on medium instances
Scalability	Solvable within 60s for $\ V\ \leq 100$
Validity	100% constraint satisfaction on all test cases

Phase	Task	Duration
Week 1	Graph pre-processing implementation (ASAP/ALAP, transitive closure)	1 week
Week 2	ILP construction + solver integration	1 week
Week 3	Testing, evaluation, visualization	1 week
Week 4	Report writing + presentation	1 week



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谢谢大家！

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